

Full-Stack & Career Development Program

Duration: 1 months

Goal: Equip learners to design, develop, deploy full-stack web applications and prepare for job placements

Course Outline

Module	Sessions	Topics
1. Python Basics	2	Session 1: Install Python, IDE setup (VS Code/PyCharm), print, variables, comments Session 2: Data types (int, float, str, bool), operators (arithmetic, comparison, logical)
2. Core Python Concepts	2	Session 3: Strings, input(), formatting Session 4: Conditionals (if, elif, else), nested conditions
3. Loops & Data Structures	2	Session 5: Loops (for, while, range, break, continue) Session 6: Lists, tuples (indexing, slicing, methods)
4. Advanced Python	2	Session 7: Dictionaries, sets (CRUD, methods) Session 8: Functions, arguments, return values, lambda
5. Python Utilities & OOP	2	Session 9: Modules (math, random, datetime), file handling (txt/csv) Session 10: Exception handling, OOP basics (class, init, inheritance)
6. Django Introduction	2	Session 11: Virtualenv, install Django, startproject, runserver Session 12: Project structure, apps, settings, URLs, MTV pattern
7. Django Development	2	Session 13: Views, HttpResponse, routing Session 14: Templates, DTL, inheritance, filters

Module	Sessions	Topics
8. Django Models & Admin	2	Session 15: Models, fields, migrations Session 16: Admin panel customization, ORM queries
9. E-commerce Core Features	2	Session 17: E-commerce models (Category, Product, Cart, Order) Session 18: Forms, ModelForms, validation
10. E-commerce Logic	2	Session 19: Cart logic (session-based), add/remove/update Session 20: Order creation flow, cart to order, status handling
11. Authentication & Finalization	2	Session 21: User authentication (register, login, logout) Session 22: Product listing, detail page, UI integration
12. Projects	2	Includes 2 Mini Projects